

Achmed Samuel Koroma

PhD Candidate · AI Researcher · Electrical & Electronic Engineer

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PROFILE

PhD candidate working on AI-driven, closed-loop automation for nanomaterials discovery. Research experience at the Italian National Research Council (CNR-IEIIT) in explainable and reliable AI, adversarial machine learning and perception for robotics, with a 2025 journal publication. Background in electrical and electronic engineering, solar PV and a decade of STEM teaching. Creates digital content with Canva, OBS Studio and AI tools.

EDUCATION

PhD — AI-Driven Automated Synthesis of Layered Nanomaterials · Doctoral research, Genoa, Italy In progress
Supervisor: Dr. Milena P. Arciniegas

International MSc in Engineering Technology for Strategy and Security · University of Genoa, Italy Sep 2021 – Feb 2024
Thesis: Adversarial Machine Learning Detection in Images via Uniform Manifold Approximation and Projection

Bachelor's Degree with Honours in Electrical and Electronic Engineering · Fourah Bay College, University of Sierra Leone Aug 2014 – Jul 2019

RESEARCH EXPERIENCE

PhD Candidate · AI-driven automated nanomaterials discovery Ongoing

- Closed-loop, AI-driven synthesis and optimisation of graded organic–inorganic layered nanomaterials (lead-free perovskites)
- Integrating a Tecan Fluent liquid-handling robot with computer vision, deep learning and egocentric wearable perception; Materials Zone for traceability
- Built an LLM-powered literature-mining platform (MACONDO) and structure–property correlation pipelines (RDKit, COSMO-RS)

Research Fellow · Italian National Research Council (CNR-IEIIT), Genoa Jun 2024 – May 2025
NextGenerationEU – RAISE (ECS00000035), subproject REXASI-PRO

- Explainable AI models (decision trees, SkopeRules, Anchors, LLMs) for robotics navigation and smart mobility; error control in collision scenarios
- Machine learning and XAI on an ALS dataset; reliable AI and adversarial robustness for perception
- Evaluated compression algorithms and dimensionality-reduction techniques; drafted technical deliverables

Research Intern — AI, Computer Vision & Data Analytics · CNR-IEIIT, Genoa Mar 2023 – May 2024

- Interpretable ML model predicting mortality risk on a clinical heart-failure dataset
- Deep learning detection of adversarial and anomalous patterns in images; UMAP, PCA, t-SNE and Isomap analysis

ENGINEERING & TEACHING EXPERIENCE

Electrical Engineer · Freetown Innovation Lab, Sierra Leone 2020 – 2022

- Solar PV installation, PV plant inspection, fault detection, maintenance and project coordination

Research & Teaching Assistant · Ernest Bai Koroma University of Science and Technology, Magburaka Jan 2021 – Feb 2022

- Supervised HND final reports; taught mathematics and engineering courses

Enumerator · West Africa Rice Cooperation (WARC Group), Freetown 2019 – 2020

Electrical Engineering Intern · Heidelberg Cement Group, Sierra Leone 2018 – 2019

- Electrical fault detection, repair and maintenance; assisted PLC and photovoltaic installations

Teacher & Head of Science Department · **Grace Community High School, Freetown** Oct 2011 – Dec 2021

- Physics, Chemistry, Biology and Pure & Applied Mathematics; curriculum development and school planning

PUBLICATIONS

Koroma, A. S.; Narteni, S.; Cambiaso, E.; Mongelli, M. (2025). Uniform Manifold Approximation and Projection Filtering and Explainable Artificial Intelligence to Detect Adversarial Machine Learning. *Information (MDPI), 16(8), 647 · Published 29 July 2025 · Open access*. doi.org/10.3390/info16080647

SKILLS

Programming	Python (very good), R (good), C++ (good), SQL (good), MATLAB (fair); FastAPI, Streamlit, Git
Machine learning	scikit-learn, TensorFlow/Keras, explainable AI (SkopeRules, Anchors, decision trees), adversarial ML (FGSM, Carlini–Wagner, DeepFool)
Data & vision	UMAP, PCA, t-SNE, Isomap; NumPy, Pandas, SciPy, Matplotlib, Plotly; OpenCV, YOLO; SPSS, Tableau, Power BI
Lab & science	Tecan Fluent, Materials Zone, RDKit, COSMO-RS (AMS), Navground simulator
Engineering	Solar PV design and installation, HOMER, PLC installation, mini-grids and hybrid systems, circuit design
Security	SIEM tools, Splunk, intrusion detection, penetration testing, network protocol analysis
AI & digital	Azure OpenAI and Gemini APIs (integration); Claude and ChatGPT (assistants); Canva, OBS Studio, AI image generation
Documentation	LaTeX/Overleaf, technical reports and deliverables, MS Office
Languages	English (native), Krio (native), Italian (A2)

PROFESSIONAL DEVELOPMENT

McKinsey Forward Program
McKinsey.org · Participant

Deep Learning with Keras and TensorFlow
Coursera · Jul 2025

Green Digital Skills
INCO Academy / Coursera · May 2025

Google Cybersecurity Professional Certificate
INCO Academy / Coursera · Jun 2024

Data Analysis for Industry
Brainnest (Germany) · Aug 2022

Special Technical Training: Mini-grids, Lithium Batteries and Hybrid Systems
Energising Development (EnDev) Mano River & GIZ · 2021

Computational Models of Visual Attention (CMVA 2026)
PhD course · 2026 · In progress

Google Project Management Professional Certificate
Coursera · Oct 2025

Get into Tech
INCO Academy · Jun 2025

Understanding Research Methods
University of London / Coursera · Apr 2025

Google Data Analytics Professional Certificate
INCO Academy / Coursera · May 2024

Design and Installation of Solar PV Systems; Energy Management (HOMER)
Regional Centre for Energy and Environmental Sustainability (RCEES), Ghana · 2021

IEEE ComSoc eHealth Technical Committee PhD School
IEEE Communications Society · 13–15 Oct 2025

Cloud Architectures & RESTful Services for Robotics
University of Genoa · PhD course

Full portfolio, projects and media: achmedsamuel.github.io/achmed-s-koroma